

Trading / VAR Page

Functional Specification

Overview

New page under the **Trading** section, named **VAR**. It calculates risk parameters (volatility of the relevant spread) from historical data, and uses them to flag — as a percentage of an accepted limit — how risky the current day's open position is. Two fully independent metrics: **SPOT** and **RB**. This is an **informational red-flag system**, not a blocking mechanism (decision on blocking deferred to a later iteration).

Core Principles (already agreed)

- Two independent metrics, calculated and displayed separately: **SPOT** (F1/F2 vs SDAC spread) and **RB** (SDAC vs CEN spread).
- Both SPOT and RB positions reset daily — zero at start of day, closed/flat by end of day. No carry-over.
- V1 risk calculation: simplest possible method. Volatility = std. dev. of historical spread per hour bucket (H01–H24). Daily VaR = simple sum of per-hour risk (no inter-hour correlation in V1).
- Result shown as **% of accepted limit** (e.g. "SPOT: 67% of limit"), not just binary breach/no-breach.
- Calibration step (Section A) is **manual** — system proposes, user accepts. No silent background recalculation.
- Daily check (Section B) is **manual / button-triggered**, consistent with the "Calculate VAR" button on the Trading tab.

Section A: Risk Parameter Calibration

Trigger: Manual buttons — "Recalculate SPOT Parameters" and "Recalculate RB Parameters" (same pattern as existing "Update FIX1 / FIX2" buttons).

Lookback window: Parameterized, not hardcoded. Editable control (numeric input or dropdown — 30/60/90/180 days). **Default: 90 days.**

Calculation, per hour bucket (H01–H24):

- **SPOT:** historical spread = Fixing (F1 or F2) – SDAC, over the lookback window.
- **RB:** historical spread = SDAC – CEN, over the lookback window.
- Volatility = standard deviation of each per-hour spread series.

Review & Accept workflow:

- After clicking recalculate, system displays the **proposed** 24-row volatility table — does not apply automatically.
- User reviews and clicks **"Accept"** to make it the active parameter set used in Section B.
- Until accepted, the previous parameter set remains active.
- User also sets/confirms the **Accepted Daily VaR Limit** — one PLN value for SPOT, one for RB.

Stored per calibration:

Calibration date/time · Lookback window used · Per-hour volatility table · Accepted limits (SPOT, RB)

Section B: Daily VAR Monitor

Trigger: Manual — a "Calculate VAR" button, available both on this page and on the main Trading tab.

Inputs: Today's open position per hour (long/short MW), separately for SPOT and RB (reset daily) + latest accepted volatility table from Section A.

Calculation, per hour:

$$\text{Risk_hour} = |\text{Volume_MW}| \times \sigma_{\text{hour}} \times z$$

σ_{hour} = accepted volatility for that hour · z = z-score for chosen confidence level (open item)

Daily aggregation (V1 — simplest method):

$$\text{Daily_VAR} = \sum \text{Risk_hour} \text{ (H01-H24, simple sum, no inter-hour correlation)}$$

Calculated independently for SPOT and RB.

Limit comparison:

$$\% \text{ of limit} = (\text{Daily_VAR} / \text{Accepted_Limit}) \times 100$$

Display:

- Two gauges: "SPOT: X% of limit", "RB: Y% of limit"

Band	Range
■ Green	< 70%
■ Yellow	70 – 100%
■ Red	> 100%

(Suggested starting point — confirm before building)

- Per-hour breakdown table: hour, position (MW, long/short), volatility used, risk contribution (PLN).
- Last calibration date and lookback window shown for context.

Open Items Requiring a Decision

- **Confidence level / z-score** — 95% ($z \approx 1.645$) or 99% ($z \approx 2.33$)? Directly scales the Daily VAR.
- **Color band thresholds** — 70%/100% above is a starting suggestion, not final.
- **Fixing selection for SPOT spread** — F1 only, F2 only, or selectable?
- **Accepted Limit input** — simple editable PLN field, or its own confirmation step?

Answers to Your Two Questions

Manual vs. automatic: Manual for both sections. Section A matches the "system proposes, I accept" workflow already agreed — automatic background recalculation would change risk parameters without your awareness. Section B stays button-triggered, consistent with the planned "Calculate VAR" button on the Trading tab.

Hardcoded vs. parameterized lookback window: Parameterized, default 90 days. Low implementation cost (just a date-range input), and the flexibility is genuinely useful early on — you'll want to compare 30/60/90/180-day windows to sanity-check the model before trusting its output.